Hotel Booking Cancellation Prediction System

Project report based on the uploaded project brief and hotel_bookings.csv

1. Problem statement and objective

Hotel booking cancellations can cause revenue loss, inefficient room allocation and difficulty in demand forecasting. The objective is to predict whether a booking will be canceled (is_canceled = 1) or not canceled (0).

2. Dataset overview

Item	Observed value
Original dataset	119,390 rows × 32 columns
Target	is_canceled: 0 = not canceled; 1 = canceled
Original target split	75,166 not canceled (62.96%); 44,224 canceled (37.04%)
Exact duplicate rows	31,994 removed
Cleaned dataset	87,396 rows × 32 columns
Target after cleaning	63,371 not canceled (72.51%); 24,025 canceled (27.49%)

The dataset contains numerical and categorical booking/customer information, including hotel, lead time, arrival date, guest counts, country, meal, market segment, deposit type, room type and booking history.

3. Data inspection and cleaning

Finding after duplicate removal	Action
company: 93.98% missing	Kept as an identifier category; missing values set to 'Unknown'.
agent: 13.95% missing	Kept as an identifier category; missing values set to 'Unknown'.
country: 0.52% missing	Missing values set to 'Unknown'.
children: 4 missing values	Filled with median (0).
ADR: one negative value	Replaced with median of valid non-negative ADR values (98.10).
Duplicate rows / column names	Removed exact duplicates; standardized names to lowercase snake_case.
Feature engineering	Added total_guests and total_stay_nights.

4. Exploratory data analysis

After duplicate removal, 27.49% of bookings are canceled and 72.51% are not canceled. The observed cancellation rate is 30.0% for City Hotel and 23.5% for Resort Hotel. By deposit type, cancellation rates are approximately 94.7% for Non Refund, 26.7% for No Deposit and 24.3% for Refundable. These are descriptive associations, not proof of causation.

5. Outlier detection and treatment

IQR checks were performed for lead_time, adr, days_in_waiting_list and stays_in_week_nights. Extreme values were not automatically deleted because they may represent genuine bookings. Right-skewed non-negative variables lead_time, adr and days_in_waiting_list were transformed using log1p. The single negative ADR value was corrected.

6–8. Encoding, scaling and train-test split

Nominal categories are encoded using OneHotEncoder(handle_unknown='ignore'). Numerical columns are median-imputed and standardized using StandardScaler. These transformations are inside a scikit-learn Pipeline and are fitted only on training data. An 80:20 stratified split (random_state=42) creates 69,916 training rows and 17,480 test rows.

Leakage prevention

reservation_status is excluded because it directly reveals the outcome (for example, Canceled or Check-Out). reservation_status_date is excluded because it is tied to the final outcome, and assigned_room_type is excluded because it may reflect a later assignment. This avoids giving the model information that would not reliably be available at the original prediction point.

9. Baseline classification model (additional project step)

Metric	Test result
Accuracy	78.04%
Precision	56.93%
Recall	82.66%
F1-score	67.42%
ROC-AUC	0.8701
Confusion matrix	[[9,670, 3,005], [833, 3,972]]

A class-weighted Logistic Regression model was added as a baseline. Its recall is relatively high, so it detects many canceled bookings; its lower precision means some bookings flagged as cancellations are not canceled. The decision threshold and business costs should be considered before operational use.

10. Conclusion and next steps

The requested workflow has been completed through train-test splitting, with a baseline model added as an extension. Further work can compare Decision Tree, Random Forest and Gradient Boosting models, tune thresholds, inspect feature importance and validate performance on newer bookings before deployment.

How to run

```
pip install pandas numpy matplotlib seaborn scikit-learn joblib
python hotel_cancellation_project.py --data hotel_bookings.csv
```

The script creates hotel_project_output/ with EDA plots, an outlier summary, model metrics and the saved model pipeline.